

02 — Data Curation, Ethics & Tokenization

Build the data pipeline correctly before training a model

Build guide for the Context-Aware Online Distress Severity Detection project.

1. Phase 0 — Define the Dataset Contract

Before training, write down the exact meaning of every label, how examples are annotated, what data sources are permitted, and what information must never be retained.

Define train/validation/test rules before experimenting with models. Prevent the test set from influencing model selection.

Document annotation disagreement and ambiguous examples instead of pretending every label is objectively obvious.

2. Phase 1 — Corpus Curation

Aggregate an ethically usable collection of anonymized posts or use an existing dataset whose license and provenance permit the intended research use.

Create a reproducible ingestion script so that the dataset can be rebuilt rather than manually copied into notebooks.

Recommended logical schema: id, text, label, source, split, and only the minimum metadata needed for research.

Avoid retaining usernames, exact locations, profile URLs, account IDs, or other unnecessary identifiers.

3. Ethical Sanitization Pipeline

Raw text should pass through sanitization before tokenization or embedding generation.

Implement regex-based and rule-based detection for usernames, URLs, emails, phone-like strings, locations where feasible, names where feasible, and unique handles.

Use hashing only when a stable pseudonymous identifier is genuinely necessary. Hashing is not the same thing as removing sensitive information.

Keep a clear record of what was transformed, removed, or retained.

4. Sanitization Flow

```
Raw Dataset
↓
PII / identifier detection
↓
Remove or replace sensitive spans
↓
Validate that no prohibited fields remain
↓
Store sanitized corpus
↓
Only then tokenize / embed
```

5. Tools for the First Version

Python for the pipeline.

Regex for deterministic pattern removal.

Pandas for tabular manipulation and validation.

Hashing/de-identification utilities where justified.

Hugging Face Datasets can later make dataset versioning and transformation cleaner.

6. Phase 2 — Tokenization

Start with the tokenizer associated with the selected pretrained transformer. Do not modify a pretrained tokenizer without an experiment demonstrating why it is needed.

RoBERTa uses subword tokenization, which helps with misspellings, slang fragments, and rare words.

If the corpus contains unusually large amounts of online vernacular, investigate domain tokenizer adaptation as a separate experiment rather than assuming it helps.

7. Data Analysis Before Training

Measure class distribution.

Inspect text length distributions.

Inspect duplicate and near-duplicate examples.

Check for train/test leakage.

Review representative examples from every class.

Identify annotation ambiguity and potentially problematic samples.

Record every preprocessing decision in code and documentation.

8. Recommended Repository Data Layout

```
data/  
  raw/  
  interim/  
  processed/  
  README.md
```

Raw data should be access-controlled and should not be committed to a public Git repository when its terms or sensitivity prohibit it.

Processed public artifacts should contain only what is safe and necessary.